SEPHORA BEAUTY E-COMMERCE DATABASE

Product Management – SQL Queries and Business Reports

1. Introduction

This document adapts the Product Management sample for the Sephora Beauty E-Commerce Database. It covers product queries, filtering, sorting, price searches, customer information, and business reports.

2. Database Tables

Table	Purpose
Category	Stores beauty category names and descriptions.
Products	Stores product name, brand, category, price, shade, and stock.
Customers	Stores basic customer information.
Orders	Stores customer, order date, total amount, and order status.
Order_Details	Stores products and quantities included in each order.

3. SQL Queries Included

3.1 Basic SQL Queries

Display all products

```
SELECT * FROM Products;
```

Display selected product details

```
SELECT Product_ID, Product_Name, Brand, Price, Stock FROM Products;
```

Filter by category (1 = Skincare)

```
SELECT Product_ID, Product_Name FROM Products WHERE Category_ID = 1;
```

Sort by price descending

```
SELECT Product_ID, Product_Name, Price FROM Products ORDER BY Price DESC;
```

Display unique brands

```
SELECT DISTINCT Brand FROM Products;
```

Display unique shades

```
SELECT DISTINCT Shade FROM Products;
```

3.2 Search Products Based on Price

Products priced between 2000 and 3500

```
SELECT Product_ID, Product_Name, Price FROM Products
WHERE Price BETWEEN 2000 AND 3500 ORDER BY Price;
```

3.3 Retrieve Customer Information

Display all customers

```
SELECT * FROM Customers;
```

3.4 Apply Filtering Conditions

AND condition

```
SELECT Product_ID, Product_Name, Price, Stock FROM Products
WHERE Price < 3500 AND Stock > 25;
```

OR condition

```
SELECT Product_ID, Product_Name, Category_ID FROM Products
WHERE Category_ID = 1 OR Category_ID = 2;
```

NOT condition

```
SELECT Product_ID, Product_Name, Shade FROM Products
WHERE NOT Shade = 'Clear';
```

3.5 Basic Business Reports

Product price report

```
SELECT Product_ID, Product_Name, Price FROM Products ORDER BY Price DESC;
```

Customer order report

```
SELECT o.Order_ID, c.Customer_Name, o.Order_Date, o.Total_Amount, o.Order_Status
FROM Orders o JOIN Customers c ON c.Customer_ID = o.Customer_ID
ORDER BY o.Order_ID;
```

Product order report

```
SELECT p.Product_ID, p.Product_Name,
       SUM(od.Quantity) AS Total_Quantity_Ordered,
       SUM(od.Quantity * od.Price) AS Total_Sales
FROM Products p JOIN Order_Details od ON od.Product_ID = p.Product_ID
GROUP BY p.Product_ID, p.Product_Name
ORDER BY Total_Sales DESC;
```

Customer-wise total orders

```
SELECT c.Customer_ID, c.Customer_Name, COUNT(o.Order_ID) AS Total_Orders
FROM Customers c LEFT JOIN Orders o ON o.Customer_ID = c.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Name
ORDER BY c.Customer_ID;
```

Customer-wise total purchase amount

```
SELECT c.Customer_ID, c.Customer_Name,
       COALESCE(SUM(o.Total_Amount), 0) AS Total_Purchase_Amount
FROM Customers c LEFT JOIN Orders o ON o.Customer_ID = c.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Name
ORDER BY Total_Purchase_Amount DESC;
```

4. SQL Concepts Used

CREATE DATABASE, CREATE TABLE, INSERT, SELECT, WHERE, BETWEEN, AND, OR, NOT, ORDER BY, DISTINCT, JOIN, LEFT JOIN, SUM(), COUNT(), GROUP BY, COALESCE(), PRIMARY KEY, and FOREIGN KEY.

5. How to Run

Open Sephora_Product_Management.sql in MySQL Workbench, connect to your MySQL server, and click the lightning-bolt Execute button. The script creates the database and sample tables, inserts sample records, and runs all queries. It drops and recreates these five sample tables, so do not use it where existing data in those tables must be preserved.

6. Conclusion

These queries help the Sephora Beauty database display and search beauty products, retrieve customer information, filter and sort records, and generate order and purchase reports.